2. Cardinality, countability, and non-negative sums

REAL VARIABLES I

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Measure theory repeatedly combines countably many sets and nonnegative quantities. Cardinality tells us which collections can be arranged in a sequence, while extended nonnegative sums let us add a family without first choosing an ordering. Together, these ideas supply the mechanism behind countable additivity and the model convergence theorems used later for integrals. While arbitrary nonnegative sums are not strictly necessary for developing Lebesgue integration theory, they are a very useful test case for dealing with "generalized sums" beyond the finite algebraic setting.
After studying these notes, you will be able to do the following.

- Compare cardinalities and recognize finite, countable, and uncountable sets.
- Prove that standard unions, products, and images of countable sets are countable.
- Define and estimate nonnegative sums over arbitrary index sets.
- Interchange nonnegative sums and apply monotone and dominated convergence to families of numbers.

2.1 Cardinality and countability

Definition 1 (Finite, infinite, and countable sets). A set A is *finite* if there are $n \in \mathbb{N}$ and a bijection

$$\iota\colon \{k \in \mathbb{N} : k < n\} \longrightarrow A.$$

For $n = 0$, the domain is empty, so this definition includes the empty set. It is *infinite* if it is not finite. A set A is *countably infinite* if there is a bijection $\iota : \mathbb{N} \longrightarrow A$. It is *countable*, or *at most countable*, if it is finite or countably infinite.

2.1.1 Comparing cardinalities

Definition 2 (Comparison of cardinalities). Two sets A and B have the same cardinality, written $A \sim B$ or $|A| = |B|$, if there is a bijection $A \to B$.
Write $|A| \leq |B|$ if there is an injection $A \to B$. Write $|A| \geq |B|$ if $|B| \leq |A|$, equivalently, if there is an injection $B \to A$.

Cardinal comparison is reflexive and transitive. The next theorem supplies antisymmetry up to bijection, which is the relevant notion of equality for cardinalities.

Theorem (Cantor–Bernstein theorem). Let A and B be sets. If there are injections $f\colon A \longrightarrow B$ and $g\colon B \longrightarrow A$, then there is a bijection $A \to B$.

The proof is outside the scope of these notes.

2.1.2 Countable sets

We are mainly concerned with at-most-countable sets. In the usual set-theoretic setting with the axiom of choice, countably infinite sets have the smallest infinite cardinality.
In practice, we record the following useful facts.

Proposition ($|\mathbb{N}|$ is the smallest infinite cardinality). If a set A is infinite, then there exists an injection $\iota : \mathbb{N} \to A$, i.e. $|\mathbb{N}| \leq |A|$.

Proposition (Characterizations of countability). For a set A, the following are equivalent:

1. A is countable;
2. either $A = \emptyset$, or there exists a countable B and a surjection $B \to A$.
3. there exists a countable B and an injection $A \to B$;

Corollary 3 (Countable unions and finite products of countable sets). Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n\in\mathbb{N}} A_n$ is countable.
Let $m \in \mathbb{N}$, and let $A_1, \dots, A_m$ be countable sets. Then $A_1 \times \cdots \times A_m$ is countable.

Proof of Corollary 3. The proof encodes pairs of natural numbers and then uses that encoding for unions and products.

Step 1: encode pairs of natural numbers.

The map

$$\pi\colon \mathbb{N} \times \mathbb{N} \longrightarrow \mathbb{N} \setminus \{0\}, \qquad \pi(r, s) = 2^r(2s + 1),$$

is a bijection by unique factorization into a power of 2 and an odd number. Hence $\mathbb{N} \times \mathbb{N}$ is countable.

Step 2: encode a countable union.

Put $A = \bigcup_{n\in\mathbb{N}} A_n$. If $A = \emptyset$, the conclusion is immediate. Otherwise, fix $a_* \in A$. For every nonempty A_n, choose a surjection $\iota_n\colon \mathbb{N} \to A_n$; for an empty A_n, define ι_n to be the constant map with value a_*. This simultaneous choice is the countable-choice step in the proof. The map

$$(n, m) \longmapsto \iota_n(m)$$

is a surjection from $\mathbb{N} \times \mathbb{N}$ onto A, so A is countable.

Step 3: encode finite products.

If either of two countable sets A and B is empty, their product is empty. Otherwise, choose surjections

$$\alpha\colon \mathbb{N} \to A, \qquad \beta\colon \mathbb{N} \to B.$$

Then $(n, m) \mapsto (\alpha(n), \beta(m))$ is a surjection from the countable set $\mathbb{N} \times \mathbb{N}$ onto $A \times B$. Induction gives the claim for every finite product; the empty product is a singleton. □

Corollary 4 (Countability of the rational numbers). The set $\mathbb{Q}$ is countable.

Proof of Corollary 4. The set $\mathbb{Z}$ is countably infinite, as witnessed by the enumeration

$$0, 1, -1, 2, -2, \dots .$$

Therefore $\mathbb{Z} \setminus \{0\}$ and $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ are countable. The map $(m, n) \mapsto m/n$ is a surjection from this product onto $\mathbb{Q}$, and a surjective image of a countable set is countable. □

Theorem 5 (Uncountability of binary sequences). The set $\{0,1\}^{\mathbb{N}} = \{(x_n)_{n\in\mathbb{N}} : x_n \in \{0,1\}\}$ is uncountable.

Proof of Theorem 5. Suppose that $\{0,1\}^{\mathbb{N}}$ is countable. Since it is nonempty, there exists a surjection $F\colon \mathbb{N} \longrightarrow \{0,1\}^{\mathbb{N}}$. Write $F(k) = (x_n^{(k)})_{n\in\mathbb{N}}$. Define $y = (y_n)_{n\in\mathbb{N}}$ by $y_n = 1 - x_n^{(n)}$. Then $y \in \{0,1\}^{\mathbb{N}}$, but $y \neq F(k)$ for every k, since their k-th coordinates differ. This contradicts surjectivity of F. □

Corollary 6 (Uncountability of the real numbers). The set $\mathbb{R}$ is uncountable.

Proof of Corollary 6. Define the base 3 expansion operation $\Phi\colon \{0,1\}^{\mathbb{N}} \longrightarrow [0,1]$ by

$$\Phi(x) = \sum_{n=0}^{\infty} \frac{x_n}{3^{n+1}}.$$

This is the ternary expansion obtained by using only the digits 0 and 1. This map is injective. Indeed, suppose $x \neq y$ and let n_0 be the first digit where they differ. By symmetry suppose $x_{n_0} = 1$ and $y_{n_0} = 0$. Since $|x_k - y_k| \leq 1$, we have

$$\begin{aligned}\Phi(x) - \Phi(y) &= \frac{x_{n_0} - y_{n_0}}{3^{n_0+1}} + \sum_{k=n_0+1}^{\infty} \frac{x_k - y_k}{3^{k+1}} \geq \frac{1}{3^{n_0+1}} - \sum_{k=n_0+1}^{\infty} \frac{|x_k - y_k|}{3^{k+1}} \\ &\geq \frac{1}{3^{n_0+1}} - \sum_{k=n_0+1}^{\infty} \frac{1}{3^{k+1}} = \frac{1}{2\cdot 3^{n_0+1}} > 0.\end{aligned}$$

The uncountability of $\{0,1\}^{\mathbb{N}}$ from Theorem 5 implies that $[0,1]$, and therefore $\mathbb{R}$, is uncountable. □

2.2 Extended reals and nonnegative sums

Definition 7 (Extended real numbers). The extended real numbers are $\bar{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, ordered by $-\infty < x < +\infty$ for every $x \in \mathbb{R}$. The extended nonnegative real numbers are $\bar{\mathbb{R}}_+ = [0, +\infty]$.
Addition and multiplication on $\bar{\mathbb{R}}_+$ extend the usual operations by
$a + (+\infty) = (+\infty) + a = +\infty$ for $a \in [0, +\infty]$, with the convention
$a \cdot (+\infty) = (+\infty) \cdot a = +\infty$ for $a \in (0, +\infty]$,

$$0 \cdot (+\infty) = (+\infty) \cdot 0 = 0.$$

Other operations are not defined.

Every subset of $\bar{\mathbb{R}}$ has a supremum and an infimum in $\bar{\mathbb{R}}$, with

$$\sup \emptyset = -\infty, \qquad \inf \emptyset = +\infty.$$

Warning (Undefined extended-real subtraction). The expression $+\infty + (-\infty)$ is undefined. An argument involving extended-real quantities may use subtraction only after the relevant terms have been shown to be finite or the expression has otherwise been given a separate definition.

2.2.1 Arbitrary nonnegative sums

Definition 8 (Arbitrary nonnegative sum). Let $\mathcal{J}$ be any index set, and let $(a_j)_{j\in\mathcal{J}}$ be a family in $\bar{\mathbb{R}}_+$. Define

$$\sum_{j\in\mathcal{J}} a_j := \sup\left\{\sum_{j\in F} a_j : F\subseteq\mathcal{J} \text{ is finite}\right\}.$$

The sum over the empty set is 0. For $K\subseteq\mathcal{J}$, the notation $\sum_{j\in K} a_j$ denotes the sum of the subfamily $(a_j)_{j\in K}$, defined by the same finite-subset supremum.

The definition immediately gives the following monotonicity properties:

- if $K\subseteq L\subseteq\mathcal{J}$ then $\sum_{j\in K} a_j \le \sum_{j\in L} a_j$,
- if $0\le a_j\le b_j$ for every j then $\sum_{j\in\mathcal{J}} a_j \le \sum_{j\in\mathcal{J}} b_j$.

We will repeatedly use the following approximation property, which follows directly from the definition of sup.

- If $S=\sum_{j\in\mathcal{J}} a_j < +\infty$, then for every $\varepsilon>0$ there is a finite $F\subseteq\mathcal{J}$ such that $S-\varepsilon<\sum_{j\in F} a_j\le S$.
- If $S=+\infty$, then for every $M>0$ there is a finite $F\subseteq\mathcal{J}$ such that $\sum_{j\in F} a_j > M$.

The next theorem shows that a family with finite sum is supported on a countable set.

Theorem 9 (Countable support of a summable family). Let $(a_j)_{j\in\mathcal{J}}$ be a family in $\bar{\mathbb{R}}_+$. If $\sum_{j\in\mathcal{J}} a_j<+\infty$, then $\{j\in\mathcal{J} : a_j\neq 0\}$ is countable.

Proof of Theorem 9. The proof separates the positive terms by size and writes the support as a countable union of finite sets. For $n\in\mathbb{N}$, set $E_n=\{j\in\mathcal{J} : a_j\ge 2^{-n}\}$. Each E_n is finite because otherwise, for every $m\in\mathbb{N}$ one could choose m distinct elements of E_n, obtaining $\sum_{j\in\mathcal{J}} a_j\ge m2^{-n}$. Since m is arbitrary, the sum would be $+\infty$.
Every j with $a_j>0$ belongs to E_n for some n. Hence $\{j\in\mathcal{J} : a_j\neq 0\}=\bigcup_{n\in\mathbb{N}} E_n$, a countable union of countable sets, which is countable. □

Increasing index sets behave like increasing sequences of numbers. We write $K_n\uparrow\mathcal{J}$ when

$$K_1\subseteq K_2\subseteq\cdots \qquad \text{and} \qquad \bigcup_{n=1}^{\infty} K_n=\mathcal{J}.$$

Likewise, $x_n\uparrow z$ means that (x_n) is nondecreasing and converges to z; in that case $z=\sup_n x_n$.

Proposition 10 (Continuity under increasing index sets). Let $(a_j)_{j\in\mathcal{J}}$ be a family in $\bar{\mathbb{R}}_+$ and suppose $K_n\uparrow\mathcal{J}$. Then

$$\sum_{j\in K_n} a_j \uparrow \sum_{j\in\mathcal{J}} a_j.$$

The sets K_n need not be finite.

Proof of Proposition 10. The upper bound follows from monotonicity. For the reverse bound, a finite

subsum is eventually contained in one K_n. Set

$$S_n \coloneqq \sum_{j\in K_n} a_j, \qquad S \coloneqq \sum_{j\in\mathcal{J}} a_j.$$

By monotonicity we have $S_n \le S_{n+1} \le S$. Taking the sup of the right-hand side we have

$$\lim_n S_n = \sup_n S_n \le S.$$

It remains to show the converse inequality: $S \le \sup_n S_n$.

Step 1: the finite-sum case.

If $S < +\infty$, for every $\varepsilon > 0$ choose a finite F_ε such that $\sum_{j\in F_\varepsilon} a_j > S - \varepsilon$. Since F_ε is finite and $K_n \uparrow \mathcal{J}$, there exists n_0 such that $F_\varepsilon \subset K_n$ for all $n \ge n_0$. This shows that $S_n \ge \sum_{j\in F_\varepsilon} a_j > S - \varepsilon$ for all n large enough. Since $\varepsilon > 0$ is arbitrary, $\sup_n S_n \ge S$.

Step 2: the infinite-sum case.

If $S = +\infty$, for every $M > 0$ choose a finite F_M such that $\sum_{j\in F_M} a_j > M$. Since F_M is finite and $K_n \uparrow \mathcal{J}$, there exists n_0 such that $F_M \subset K_n$ for all $n \ge n_0$. This shows that $S_n \ge \sum_{j\in F_M} a_j > M$ for all n large enough and therefore, since M is arbitrary, $\sup_n S_n = +\infty$. □

Taking $\mathcal{J} = \mathbb{N}$ and $K_n = \{1, \dots, n\}$ in Proposition 10 gives

$$\sum_{j\in\mathbb{N}} a_j = \lim_{n\to\infty} \sum_{j=1}^{n} a_j.$$

Thus Definition 8 agrees with the usual definition of a nonnegative series.

Proposition 11 (Tonelli for nonnegative sums). Let A and B be arbitrary sets, and let $(a_{i,j})_{(i,j)\in A\times B}$ be a family in $\bar{\mathbb{R}}_+$. Then

$$\sum_{i\in A}\left(\sum_{j\in B} a_{i,j}\right) = \sum_{(i,j)\in A\times B} a_{i,j} = \sum_{j\in B}\left(\sum_{i\in A} a_{i,j}\right).$$

Proof of Proposition 11. We only prove $\sum_{i\in A}\left(\sum_{j\in B} a_{i,j}\right) = \sum_{(i,j)\in A\times B} a_{i,j}$. The other claim is symmetric.

The definition by finite subsums suggests proving each inequality by choosing a finite set of indices.

Step 1: compare a finite subset of the product.

We first show that

$$\sum_{i\in A}\left(\sum_{j\in B} a_{i,j}\right) \ge \sum_{(i,j)\in A\times B} a_{i,j}.$$

Indeed, if $F \subseteq A \times B$ is finite, then $F \subseteq \pi_A(F) \times \pi_B(F)$, where $\pi_A(F)$ is the projection of F onto A, and $\pi_B(F)$ is defined analogously. Hence

$$\sum_{(i,j)\in F} a_{i,j} \le \sum_{i\in\pi_A(F)}\sum_{j\in\pi_B(F)} a_{i,j} \le \sum_{i\in\pi_A(F)}\sum_{j\in B} a_{i,j} \le \sum_{i\in A}\sum_{j\in B} a_{i,j}.$$

Taking the supremum over finite $F \subseteq A \times B$ gives the first inequality.

Step 2: approximate the iterated sum by finitely many entries.

For the reverse inequality, set $T \coloneqq \sum_{(i,j)\in A\times B} a_{i,j}$. If $T = \infty$, the desired reverse inequality is immediate. Suppose $T < \infty$ and, toward a contradiction, that $\sum_{i\in A}\sum_{j\in B} a_{i,j} > T$. Then there is a finite set $I \subseteq A$ such that $\sum_{i\in I}\sum_{j\in B} a_{i,j} > T$. Since I is finite, for each $i \in I$ one may choose a finite set $J_i \subseteq B$ so that $\sum_{i\in I}\sum_{j\in J_i} a_{i,j} > T$. Setting $J = \bigcup_{i\in I} J_i$, we obtain

$$T \geq \sum_{(i,j)\in I\times J} a_{i,j} \geq \sum_{i\in I}\sum_{j\in J_i} a_{i,j} > T,$$

a contradiction. This proves the first equality; interchanging A and B proves the second. □

2.2.2 Convergence theorems for nonnegative sums

Proposition 12 (Monotone convergence for nonnegative sums). Let $\mathcal{J}$ be an arbitrary index set. For $j \in \mathcal{J}$ and $k \in \mathbb{N}$, let $a_{j,k} \in \bar{\mathbb{R}}_+$. Suppose that, for every $j \in \mathcal{J}$,

$$a_{j,k} \uparrow a_j \qquad \text{as } k \to \infty.$$

Then

$$\sum_{j\in\mathcal{J}} a_{j,k} \uparrow \sum_{j\in\mathcal{J}} a_j.$$

Proof of Proposition 12. Left as exercise □

Proposition 13 (Dominated convergence for nonnegative sums). Let $\mathcal{J}$ be an arbitrary index set, and let $(a_{j,k})_{(j,k)\in\mathcal{J}\times\mathbb{N}}$ be a family in $\bar{\mathbb{R}}_+$. Suppose that

- $a_{j,k} \to a_j$ as $k \to \infty$ for each $j \in \mathcal{J}$;
- there is a dominating family $(g_j)_{j\in\mathcal{J}}$ in $\bar{\mathbb{R}}_+$ such that $\sum_{j\in\mathcal{J}} g_j < +\infty$ and

$$a_{j,k} \leq g_j \qquad \text{for every } j \in \mathcal{J} \text{ and } k \in \mathbb{N}.$$

Then

$$\lim_{k\to\infty} \sum_{j\in\mathcal{J}} a_{j,k} = \sum_{j\in\mathcal{J}} a_j.$$

Proof of Proposition 13. Left as exercise □